Product Inventory Management System in Salesforce

Aim

Develop an application in Salesforce using Apex Programming and Visualforce Page to manage Product Inventory records.

Fields Used:

- Product Name
- Serial No
- Manufacture Date
- Expiry Date

Step 1: Create Salesforce Developer Account

Open Salesforce Developer Signup page: https://developer.salesforce.com/signup

Create your developer account and login.

Step 2: Open Developer Console

After login:

1. Click on Setup (⚙️)
2. Select Developer Console

Developer Console will open in a new window.

Step 3: Create Custom Object

Go to:

Setup → Object Manager → Create → Custom Object

Fill the following details:

Property	Value
Label	Product Inventory
Plural Label	Product Inventories
Object Name	Product_Inventory
Record Name	Product Name
Data Type	Text

Click Save.

Step 4: Create Custom Fields

Open Product Inventory Object.

Go to: Fields & Relationships → New

Create the following fields.

Field 1: Serial No

Property	Value
Data Type	Text
Field Label	Serial No
Length	50

Click Next → Save.

Field 2: Manufacture Date

Property	Value
Data Type	Date
Field Label	Manufacture Date

Click Next → Save.

Field 3: Expiry Date

Property	Value
Data Type	Date
Field Label	Expiry Date

Click Next → Save.

Step 5: Create Apex Controller Class

Open: Developer Console → File → New → Apex Class

Class Name:

ProductInventoryController

Paste the following code:

```
public class ProductInventoryController {

    public Product_Inventory__c productObj {get; set;}

    public ProductInventoryController() {
        productObj = new Product_Inventory__c();
    }

    // Save Product Record
    public PageReference saveProduct() {

        insert productObj;

        ApexPages.addMessage(
            new ApexPages.Message(
                ApexPages.Severity.CONFIRM,
                'Product Record Saved Successfully!'
            )
        );

        productObj = new Product_Inventory__c();

        return null;
    }
```

```
    // Fetch Product Records
    public List<Product_Inventory__c> getProductList() {

        return [
            SELECT Id,
                   Name,
                   Serial_No__c,
                   Manufacture_Date__c,
                   Expiry_Date__c
            FROM Product_Inventory__c
            ORDER BY CreatedDate DESC
        ];
    }
}
```

Click File → Save.

Step 6: Create Visualforce Page

Open: Developer Console → File → New → Visualforce Page

Page Name:

ProductInventoryPage

Paste the following code:

```
<apex:page controller="ProductInventoryController">

    <apex:form>

        <apex:pageBlock title="Product Inventory Management System">

            <apex:pageMessages />

            <apex:pageBlockSection columns="1">

                <apex:inputField value="{!productObj.Name}" />

                <apex:inputField value="{!productObj.Serial_No__c}" />

                <apex:inputField value="{!productObj.Manufacture_Date__c}" />
```

```
                <apex:inputField value="{!productObj.Expiry_Date__c}" />

            </apex:pageBlockSection>

            <apex:commandButton
                value="Save Product"
                action="{!saveProduct}"
            />

        </apex:pageBlock>

        <br/>

        <apex:pageBlock title="Product Records">

            <apex:pageBlockTable value="{!productList}" var="p">

                <apex:column value="{!p.Name}" headerValue="Product Name"/>

                <apex:column value="{!p.Serial_No__c}" headerValue="Serial No"/>

                <apex:column value="{!p.Manufacture_Date__c}"
headerValue="Manufacture Date"/>

                <apex:column value="{!p.Expiry_Date__c}" headerValue="Expiry
Date"/>

            </apex:pageBlockTable>

        </apex:pageBlock>

    </apex:form>

</apex:page>
```

Click File → Save.

Step 7: Run the Visualforce Page

Go to:

Setup → Visualforce Pages

Open: ProductInventoryPage

Click Preview.

Step 8: Test the Application

Enter:

- Product Name
- Serial No
- Manufacture Date
- Expiry Date

Click: Save Product

The record will:

- Save into Salesforce database
- Display in the table below

Expected Output

The application should:

- Insert Product Inventory records
- Display all saved records
- Use Apex Controller
- Use Visualforce Page
- Store records in Custom Object

Important Salesforce Components Used

Custom Object

Used to store Product Inventory data.

Apex Class

Used for backend business logic.

Visualforce Page

Used to create frontend user interface.

SOQL

Used to fetch records from Salesforce database.

Viva Questions and Answers

1. What is Apex?

Apex is Salesforce backend programming language used for business logic.

2. What is Visualforce?

Visualforce is a framework used to design custom Salesforce pages.

3. What is a Custom Object?

A Custom Object is a user-created database table in Salesforce.

4. What is SOQL?

SOQL stands for Salesforce Object Query Language used to retrieve records.

5. What is the use of Controller?

Controller handles application logic and database operations.

6. Why do we use Visualforce Pages?

To create custom UI in Salesforce.

7. What does insert keyword do in Apex?

It saves records into Salesforce database.

Conclusion

Successfully developed a Product Inventory Management System in Salesforce using Apex and Visualforce Page to manage product records efficiently.